

On the Influence of Shape Factors for CHF Predictions with the Sub Channel Code SUBCHANFLOW during a Rod Ejection

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ABSTRACT

Reactivity initiated transients are evaluated with reactor dynamic codes such as DYN3D using either own thermal hydraulic modules or coupled with best-estimate system codes like ATHLET, or RELAP5. The main disadvantage of such approaches is that they do not permit the direct prediction of local safety parameters like MDNB both the neutron physical and thermal hydraulic models are solved in an approximate manner at fuel assembly level. Therefore, performing safety evaluations in this context implies the use of hot channel factors. For the neutron physical part, the introduction of pin power reconstruction (PPR) methodologies in nodal methods is the first step that allows the prediction of the pin power within a fuel assembly. With this PPR capability, the pin power distribution of selected fuel assemblies can be predicted. In this work, an off-line coupling of DYN3D with SUBCHANFLOW is used for the determination of the pin power of a selected fuel assembly during a rod ejection accident (REA). The main focus of this paper is the analysis of the critical heat flux (CHF) correlations used in SUBCHANFLOW and the influence of shape or form factors, which account for non-uniform axial heat flux profiles, cladding temperature, void and other relevant parameters. The time dependent pin power distribution, predicted by DYN3D using the pin power reconstruction method, is extracted and transferred to SUBCHANFLOW. The problem investigated consists of a 3x3 fuel assembly mini core, where the fuel assemblies contain fresh fuel pins. The REA leads to a sudden power increase for a short time which is stopped by the Doppler reactivity. In such situations, safety margins may be undershot and hence an evaluation is necessary. In this analysis, CHF correlations and different form factors were selected. The main advantage of this approach is that at each axial position of each pin major safety parameters can be calculated and evaluated using a sub-channel code. The results obtained with SUBCHANFLOW show that the power peak causes a short term boiling crisis and that the selection of the form factors has an impact on the evolution of the resulting cladding temperatures. The main differences are in the duration of the critical state – the time exceeding the critical heat flux. Another result is that the option where no form factor is applied is very similar to the cases with form factors even though the applied axial power distribution is far from being uniform.

KEYWORDS

DNB, SUBCHANFLOW, DYN3D, Rod Ejection

1. INTRODUCTION

The correct prediction of the critical heat flux (CHF) is important for the operation of nuclear reactors or experimental facilities. At certain heat fluxes, the boiling crisis can occur and the heated surface is isolated from the coolant by a vapor layer which might lead to the failure of the cladding.

In the past decades, several experiments have been performed in order to analyze the CHF at various conditions. As a result, look-up tables and empirical models have been developed to calculate the CHF as a function of the mass flux, the hydraulic and heated diameter, the pressure, the heated length and the inlet sub-cooling. For the present analysis the sub channel code SUBCHANFLOW is used which contains six different CHF models and four different form factor correction to account for non-uniform heat flux distributions in axial directions.

In the frame of the European Union project NURISP [1], the improvement, extension and integration of the pin power reconstruction method (PPR) of DYN3D [2] has been achieved. The flexibility of the new development allows local refinement in the spatial mesh for specific regions of interest or even having a whole core with pin-by-pin resolution [3]. With such extended pin power reconstruction capability a 3D time dependent power map can be generated. Thus, exporting that information to SUBCHANFLOW [4], each pin has its own power profile, doing 3D sub channel analysis.

The short term rod ejection transient, performed at hot zero power condition, introduces heat into the system as a consequence of a power peak following the rod ejection. The gained power profile does not only include the power distribution of each pin but also the power as a function of time making it a transient 3D sub channel analysis.

The REA is part of the reactivity insertion accidents and during an analysis of such accident scenarios certain criteria have to be met in order to demonstrate the effectiveness of specific safety features of the plant/installation. In the IAEA safety report series no. 30, Accident Analysis for Nuclear power Plants with Pressurized Water Reactors [5], acceptance criteria for design based accidents are listed. Concerning the cladding temperature a maximum value of 1480°C is given in the IAEA document (1200°C for LOCA) is given. Furthermore, the document state that fuel melting must not occur, in some cases a maximum of 10% is allowed during certain circumstances.

This study can be considered as one part of a series of investigations dealing with DNB constrains following a rod ejection transient. A first study was performed to evaluate the capabilities of the PPR method implemented in DYN3D [6]. With that method the local safety parameters can be calculated. During that investigation only one CHF correlation was applied to the problem since the focus was on the general validity of the methodology.

The aim of this investigation is to identify the influence of different form factors applied to several CHF correlations on the superheating of the cladding. For this purpose, a 3x3 PWR like mini core is investigated, where the central FA is modeled at pin level.

The consequences for the cladding and the fuel will not be addressed here since the intention of this paper is to proof whether 1) the applied power peak is sufficient to enter into local boiling crisis and 2) to show that SUBCHANFLOW can calculate thermo-hydraulic safety parameters during local boiling crisis.

2. DESCRIPTION OF THE MINI CORE AND THE TRANSIENT

The mini core consists of a 3x3 matrix of fresh MOX elements with a fresh UO₂ element in the central position (17x17 pins). The mini core is surrounded by a row of reflector assemblies, see Fig. 1. Such mini core has been proposed for a boron dilution benchmark [7] and is based on the OECD/NEA and U.S. NRC MOX core transient benchmark [8].

A transient calculation starting from hot zero power (HZIP) is defined. A control rod partially inserted in the central assembly will be fully ejected in 0.1 seconds leading to a very fast insertion of reactivity in the system. As a consequence, the rapid increase in reactivity brings the system to a prompt critical state and a significant power rise is expected in the initial few milliseconds. The strong negative fuel temperature (Doppler) feedback reactivity is the only inherent reactivity feedback responsible to stop the power excursion and to mitigate the power peak. The fuel temperature is a prompt and inherent reactivity feedback related to the Doppler broadening of resonances and is directly associated with the fuel temperature changes.

For the purpose of this analysis, this hypothetical transient will bring the mini core to a critical state, in terms of exceeding the critical power (MDNBR < 1.0), in which it is important to evaluate the impact of the power peak on the local safety parameters. The local perturbation justifies the intention of having mesh refinements focusing on central FA. For this mini core configuration, the soluble boron concentration is fixed to 490 pcm in order to have a critical initial steady state. The multiplication factor of the system calculated with DYN3D for the steady state is $k_{\text{eff}} = 0.999988$.

Due to features of that mini core, the models and approaches, described in the context of this paper will be evaluated for their applicability for similar and more realistic scenarios.

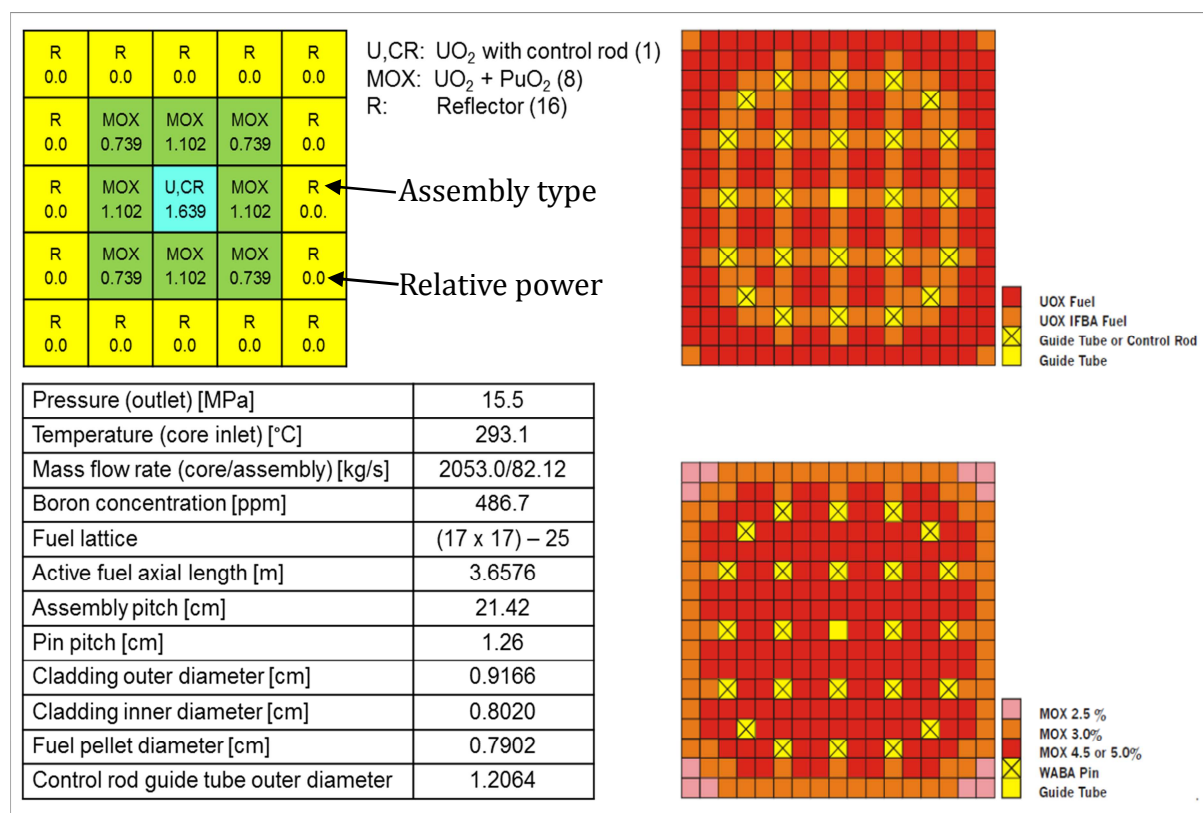


Fig. 1 Layout and operational conditions of the 3x3 PWR like mini core

3. CRITICAL HEAT FLUX CORRELATIONS AND FORM FACTORS

3.1. General remarks

For the present investigation, the incompressible equations are solved in SUBCHANFLOW. Due to the fast power insertion, very small time steps are considered and the maximum is set to 10^{-3} s while the lower limit for the weakly compressible option is given by the speed of sound of the mixture.

Due to the delayed release of the power to the coolant (heat conduction and convection in the fuel, cladding and coolant), it is justified to neglect the simulation of pressure waves caused by rapid evaporation or condensation.

In SUBCHANFLOW two parameters will be calculated to distinguish if critical heat flux is exceeded or not. The first parameter is the classical minimum departure from nucleate boiling (MDNBR), calculated as follows:

$$\text{MDNBR} = \frac{q''_{\text{crit}}}{q''_{\text{local}}} \quad (1)$$

In case the MDNBR is less than 1.0 the critical heat flux is exceeded. In other words, a number greater than 1.0 indicates the margin till the critical heat flux is reached. For the present case this ratio will not be considered due to the mentioned delay in power release and the fact that in transition and flow boiling (on the right side of the point of critical heat flux in Fig. 2) that ratio has no meaning.

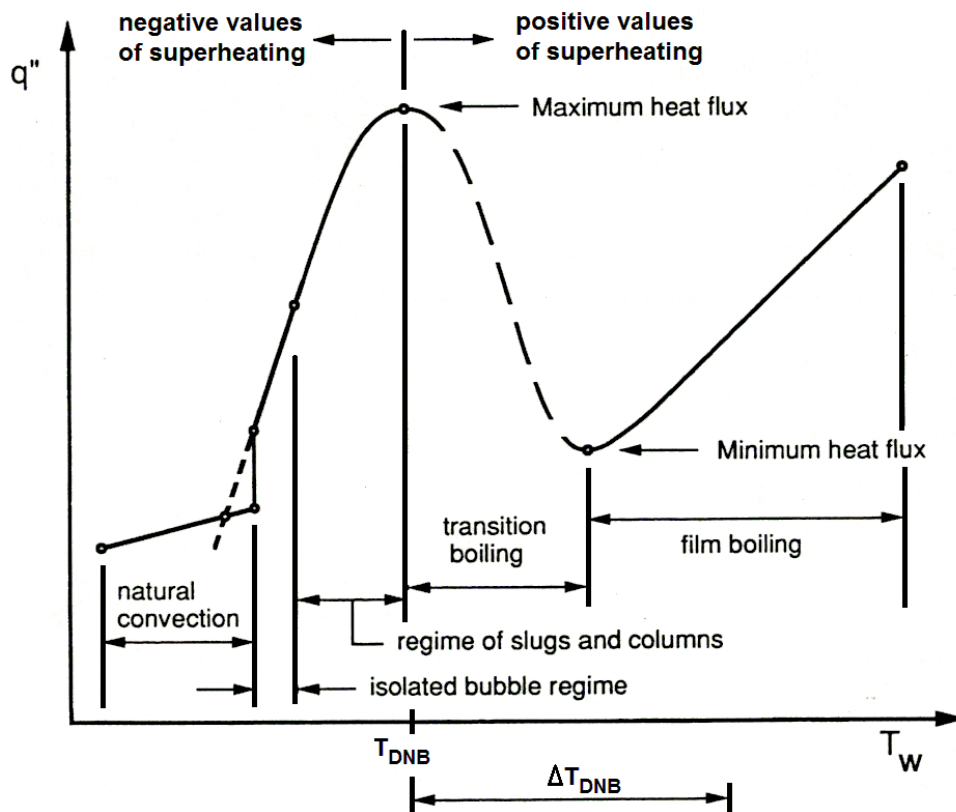


Fig. 2 Boiling curve [9] with information regarding the location of superheating

This is especially the case after the power peak. The MDNBR is calculated with the actual heat flux which might after the peak already be at a normalized level. But the superheating and the cladding temperature reach the maximal values later due to the delay of the consequences of the power peak. The local MDNBR would be greater than 1.0 but practically, local boiling crisis would still occur. Therefore, an indicator was selected which considers also the local temperatures at the same point in time. This indicator is the difference between the actual cladding temperature and the DNB temperature of the cladding at CHF (in the further context that indicator is called superheating). The correlation is given in Eq. (2).

$$\Delta T_{\text{DNB}} = t_{\text{clad}} - \left(t_{\text{cool}} + \frac{q''_{\text{crit}}}{h_{\text{crit}}} \right) \quad (2)$$

Positive values indicate that the point of critical heat flux is exceeded. While negative values indicate the margin till critical heat flux is reached.

3.2. CHF correlations

SUBCHANFLOW offers the user 6 different critical heat flux correlations/models. Among them, the models of OKB and Westinghouse – the ones used in the present investigation, Eq. (3) and (4). These models have been mostly derived from experiments with uniform heat flux profiles. In case the profile is non-uniform, as it is in a nuclear reactor, form factors have been developed in order to correct the uniform heat flux CHF correlations. Validation of critical heat flux/power capabilities of SUBCHANFLOW at steady state conditions has been proofed during internal investigations by means of NUPEC PFBT and BFBT experiments and is about to released soon.

The OKB correlation [10], Eq. (3), has been developed for VVER reactor types. In an experimental investigation by Bol'shakov et al. [11], 200 burnout regimes have been observed while maintaining the following parameter range: 7 MPa < p < 15.7 MPa; 1000 kg/m²s < G < 5500 kg/m²s; and a relative enthalpy at the inlet to the heated zone ranging from -0.49 to + 0.42. A general conclusion of Bol'shakov et al. analyses is that the correlation, given in Eq. (3), under predicts the experimental results over the entire mass flux range. That under prediction can be in the range of up to 25 %.

$$q''_{\text{crit,OKB}} = 7.95 \cdot 10^5 \cdot (1 - x)^{-0.5+1.05 \cdot 10^{-7} \cdot p} \cdot G^{-0.127+0.311 \cdot (1-x)} \cdot (1 + 1.85 \cdot 10^8 \cdot p) \quad (3)$$

The Westinghouse W-3 CHF correlation, Eq. (4), proposed by Tong [12], is one of the most known and used correlation in nuclear industry. Developed to meet the peculiarities of Westinghouse type PWR's, it has been validated for the following parameter range: 5.5 MPa < p < 15.9 MPa; 1350 kg/m²s < G < 6800 kg/m²s; 0.508 cm < d < 1.27 cm; -0.15 < x < 0.15; h > 930 kJ/kg; 24.5 cm < L < 366 cm

$$q''_{\text{crit,W3}} = \{ (2.022 - 6.238 \cdot 10^{-8} \cdot p) + (0.1722 - 1.427 \cdot 10^{-8} \cdot p) \cdot \exp[(18.177 - 5.987 \cdot 10^{-7} \cdot p) \cdot x] \} \cdot [(0.109 - 1.173 \cdot x + 0.127 \cdot x \cdot |x|) \cdot 1.0 \cdot 10^{-3} \cdot G + 3271] \cdot [1.157 - 0.869 \cdot x] \cdot [0.2664 + 0.8357 \cdot e^{-124.1 \cdot d}] \cdot [0.8258 + 3.413 \cdot 10^{-7} \cdot (h_{\text{sat}} - h_{\text{in}})] \cdot 3.1546 \cdot 10^6 \quad (4)$$

3.3. CHF form factors for non-uniform heat flux

The above presented correlations have been developed using experimental data where the heat flux was applied uniformly over the whole heated length. Therefore, corrective measures have been developed in order to account for non-uniform heat flux scenarios, as they are present in nuclear reactors. In SUBCHANFLOW five form factor options are available. The first one does not consider a correction and it is set to 1.0. The remaining four factors are based on shape factors developed for/by: COBRA-IVi [13], Tong [12], W-3 [14], and Smolin [15].

The general form of the form factor is defined as follow:

$$F = \frac{q''_{\text{crit,uniform}}}{q''_{\text{crit,non-uniform}}} \quad (5)$$

The semi-empirical model developed by Tong et al. [12] writes as follow:

$$F = \frac{C}{1 - e^{-C \cdot l_{\text{crit}}}} \cdot \int_0^{l_{\text{crit}}} \frac{q''(l)}{q''(l_{\text{crit}})} \cdot e^{-C \cdot (l_{\text{crit}} - l)} dl \quad (6)$$

where

$$C_{\text{Tong}} = 17.323 \cdot \frac{[1 - x(l_{\text{crit}})]^{7.9}}{\left(\frac{G}{1356}\right)^{1.72}} \quad (7)$$

COBRA-IVi form factor is defined as the form factor proposed by Tong. The C factor, Eq. (8), follows the work of Wilson et al. [13].

$$C_{\text{COBRA}} = 265.51 \cdot \frac{[1 - x(l_{\text{crit}})]^{7.82293}}{G^{0.45758}} \quad (8)$$

As for the COBRA-IVi model, the W-3 form factor is defined as in Eq. (6) and the C factor, Eq. (9), is based on work related to the development of the THERMIT code [14].

$$C_{\text{W3}} = 185.6 \cdot \frac{[1 - x(l_{\text{crit}})]^{4.31}}{G^{0.478}} \quad (9)$$

The approach of Smolin and Polyakov [15] is similar to the one of Tong but has a dependency on the hydraulic diameter instead of the mass flux for the C factor.

$$F = C \cdot \int_0^{l_{\text{crit}}} \frac{q''(l)}{q''(l_{\text{crit}})} \cdot e^{-C \cdot (l_{\text{crit}} - l)} dl \quad (10)$$

where

$$C_{\text{Smolin}} = \frac{1}{40 \cdot d} \quad (11)$$

Due to the definition of the form factors, three different behaviors if applied to the same

correlation can be expected. The first trend is the one for the case if no form factor is applied at all. It is interesting to see how with that “simple” model the superheating, and subsequent coolant and cladding temperature, can be predicted. The results of the Tong, the COBRA IVi and the W-3 model should be similar since the foundation of these form factors is the semi-empirical model of Tong. The only difference is the definition of the C factor, which are in fact similar to each other. Results predicted using the Smolin and Polyakov form factor should differ from the previous since the definition of the form factor is different. The model of Smolin and Polyakov is simpler as it does not employ the $1/(1-\exp(-C \cdot l))$ factor in Eq. (11).

4. SIMULATION OF THE ROD EJECTION TRANSIENT

4.1 Calculation approach

The calculation is performed at assembly base but in the central assembly a more detailed description of the power distribution has been obtained by means of the pin power reconstruction method. A flowchart of the calculation approach is shown Fig. 3. DYN3D performs the neutronic calculation at nodal level including feedbacks from its own 1D thermal-hydraulic module FLOCAL. The neutronic/thermal-hydraulic coupling between DYN3D and FLOCAL is a two-way coupling. The neutronic module provides parameters to FLOCAL module and receives the feedback. But for the data transfer between DYN3D and SUBCHANFLOW a one-way approach is used. The pin power reconstruction is done at selected times steps for FA or clusters of FAs. In the present case, the PPR method is applied for the central FA. The output data from the pin power reconstruction are elaborated by auxiliary tools, extensively described in [6] and sent to SUBCHANFLOW in form of time dependent power tables. This code performs (for the central assembly) a thermal-hydraulic transient calculation on a pin-by-pin resolution using the heat addition data from the pin power reconstruction following the sub channel centered approach ($18 \times 18 = 324$ sub channels).

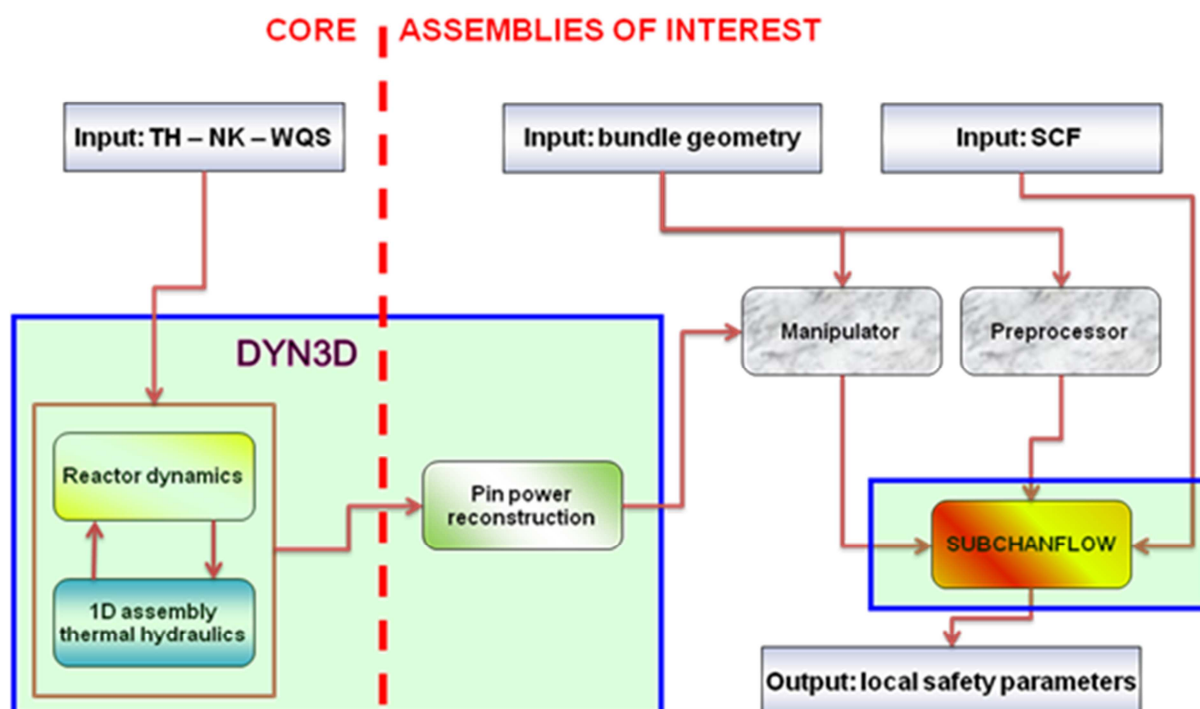


Fig. 3 Transient assembly power and relative 3D power at the time of the power peak [6]

Therefore, data are available on the heat generated by every pin in the central assemblies at every axial mesh at every time step. With the refined modeling of the central fuel assembly possible superheating can be calculated at any local level. For the present nodalization of the 17x17 FA with 20 axial levels safety parameters at 5780 positions can be calculated and evaluated. The total power as a function of time is plotted on the left side of Fig. 4. The power rises within 0.1 s from zero to around $6 \cdot 10^{10}$ W and stabilizes at a value of $5.7 \cdot 10^7$ W. The radial and axial power profile of the central assembly at the moment of the power peak is given in the right side of Fig. 4. It can be seen that the power profile is non-uniform in axial direction.

4.2. Local safety parameters predicted by SUBCHANFLOW

The maximal superheating in the central subassembly is plotted in Fig. 5 for the OKB CHF correlation and in Fig. 8 for the W-3 correlation, applying all available form factors. The sharp increase corresponds with the power peak, see Fig. 3. Right at the power peak the temperature rises immediately by 350-400°C and then the further increase is done much slower, till about 0.5 seconds before it reaches its maximum. In general, the qualitative and quantitative comparison shows a high level of agreement between the different correlations and form factors, shown in Figs. 5 – 10 and Tab. 1.

For the OKB correlation the maximum superheating for the different form factor options varies from 431°C, for the option with Smolin factor, to 436°C with no form factor. With the W-3 CHF correlation the superheating ranges from 406°C (Smolin) to 418°C (no form factor). That means that, on one side, the variation is higher for W-3 CHF correlation (12°C, while only 5°C for OKB) and, on the other side, the W-3 CHF correlation predicts a lower superheating, roughly 20°C less than the OKB correlation. That can be also seen in the resulting cladding temperatures, depicted in Fig. 6 (OKB) and Fig. 9 (W-3). For OKB correlation the range of cladding temperature is from 782 to 787°C and for W-3 it ranges from 756 to 769°C. The fuel centerline temperature (Fig. 6 and Fig. 9) is with 1504-1507°C almost constant for all options.

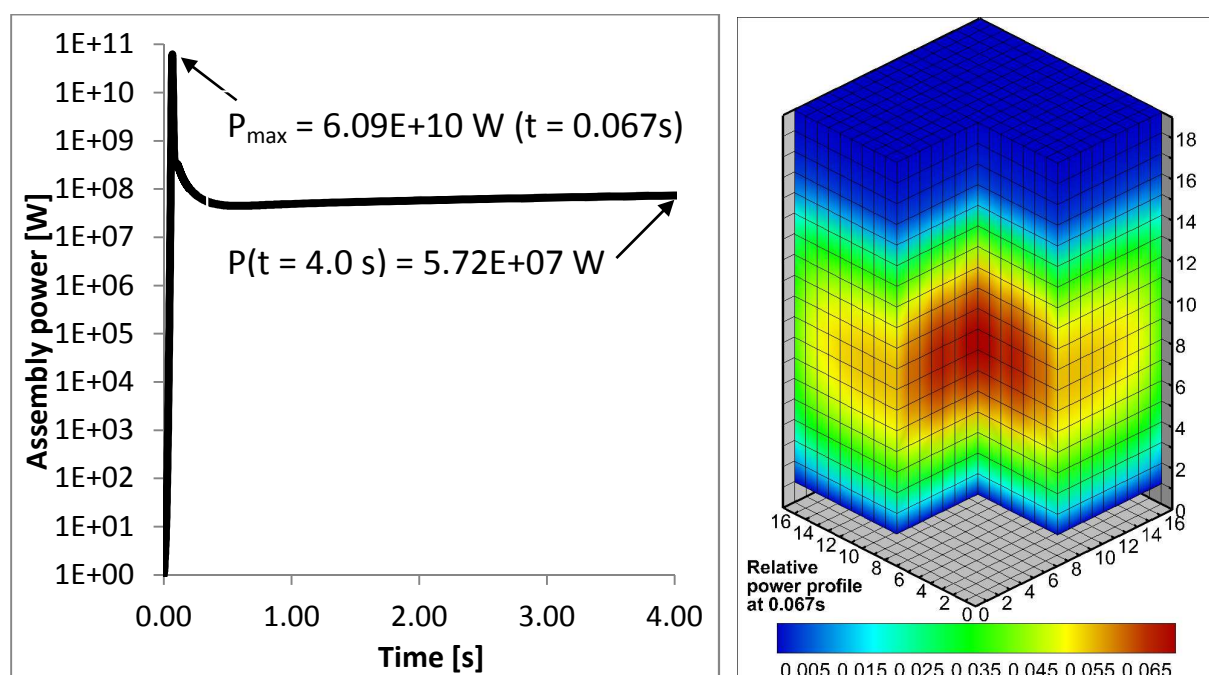


Fig. 4 Transient assembly power and relative 3D power at the time of the power peak

Table 1 Quantitative comparison of the maximal values of relevant parameters

	$\Delta T_{\text{DNB}} [^{\circ}\text{C}]$	$t_{\text{clad}} [^{\circ}\text{C}]$	$t_{\text{fuel}} [^{\circ}\text{C}]$	$\alpha [-]$	$\tau_{\text{return}} [\text{s}]$
OKB correlation					
None	436	787	1507	0.3425	3.0151
COBRA IVi	435	786	1507	0.3280	3.1551
Tong	435	786	1507	0.3235	3.2251
W-3	435	786	1507	0.3185	3.2251
Smolin	431	782	1506	0.3401	3.0651
W-3 correlation					
None	418	769	1505	0.3306	2.7951
COBRA IVi	417	768	1504	0.3106	3.0051
Tong	414	765	1504	0.3008	3.2451
W-3	414	765	1504	0.2968	3.2351
Smolin	406	756	1502	0.3046	3.3951

In addition, it can be observed that the system needs several seconds until it reaches a stable level according to the power after the power peak. The power peak is noticeable for only 0.1 seconds. However, in case of OKB, the superheating recovers to zero after 3.01 – 3.23 sec. Using the W-3 correlation, the point of recovery to stable condition is between 2.8 and 3.4 sec.

Besides the cladding and fuel centerline temperatures, the coolant temperatures can be seen in Figs. 6 and 9, too. The rise of the coolant temperature is much more delayed. After 0.3-0.4 seconds the coolant temperature reaches saturation condition. Therefore, it is worth to check local void fractions in order to see whether this power peak and the resulting superheating will cause also local boiling. The maximum void fractions are plotted in Fig. 7 and Fig. 10. In both cases, the maximum void fraction is between 0.30 and 0.34. It recovers to zero in about the same times as the superheating. At the time the void fraction is back to zero the coolant temperature decreases slightly. After the superheating returned to zero the actual power in the subassembly cause the cladding temperature to stay at values of around 340°C.

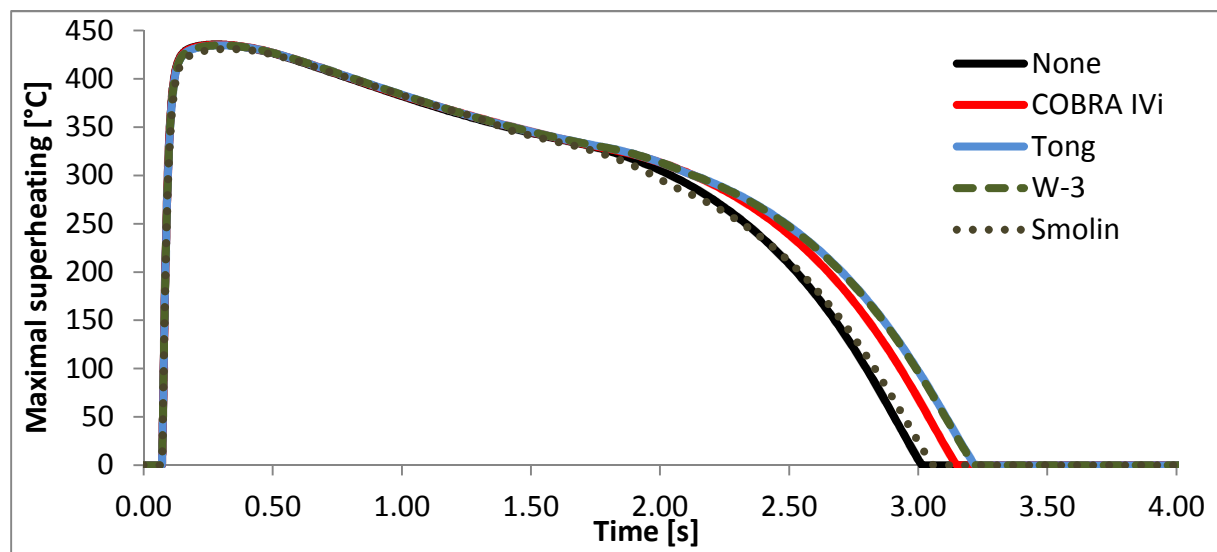


Fig. 5 Maximal values of superheating for all shapes with OKB CHF correlation

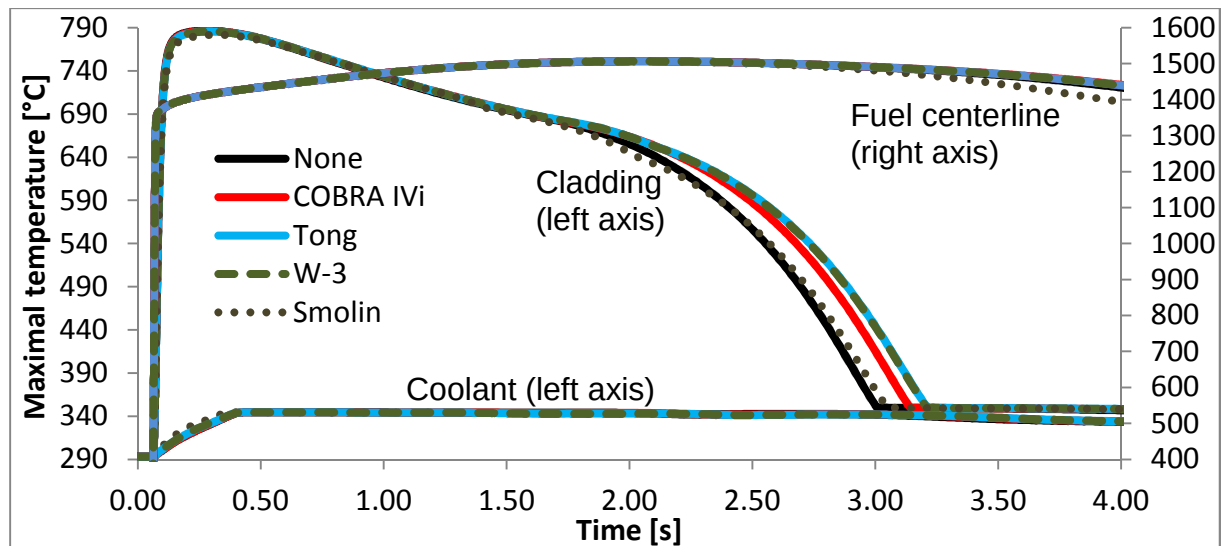


Fig. 6 Maximal values of coolant and cladding temperatures for all shapes with OKB CHF correlation

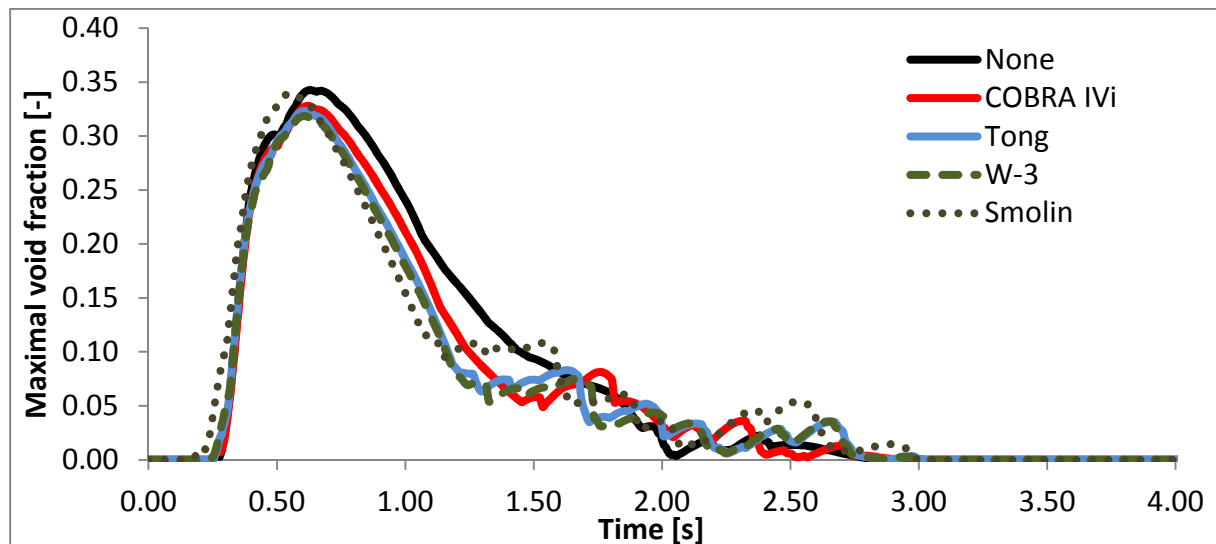


Fig. 7 Maximal values of void fraction for all shapes with OKB CHF correlation

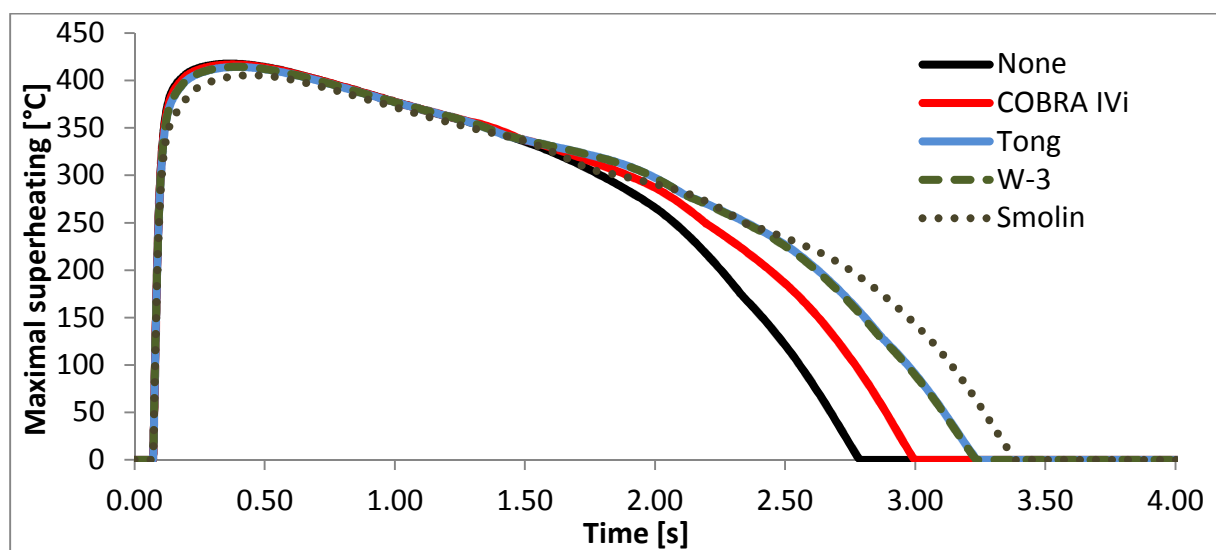


Fig. 8 Maximal values of superheating for all shapes with W-3 CHF correlation

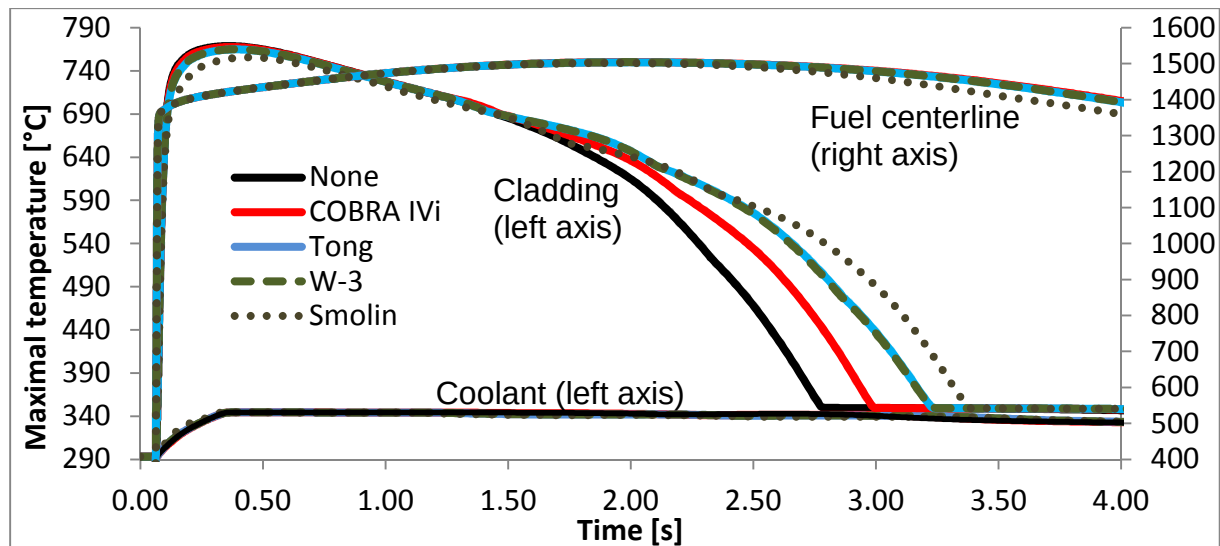


Fig. 9 Maximal values of coolant, cladding and fuel centerline temperatures for all shapes with W-3 CHF correlation

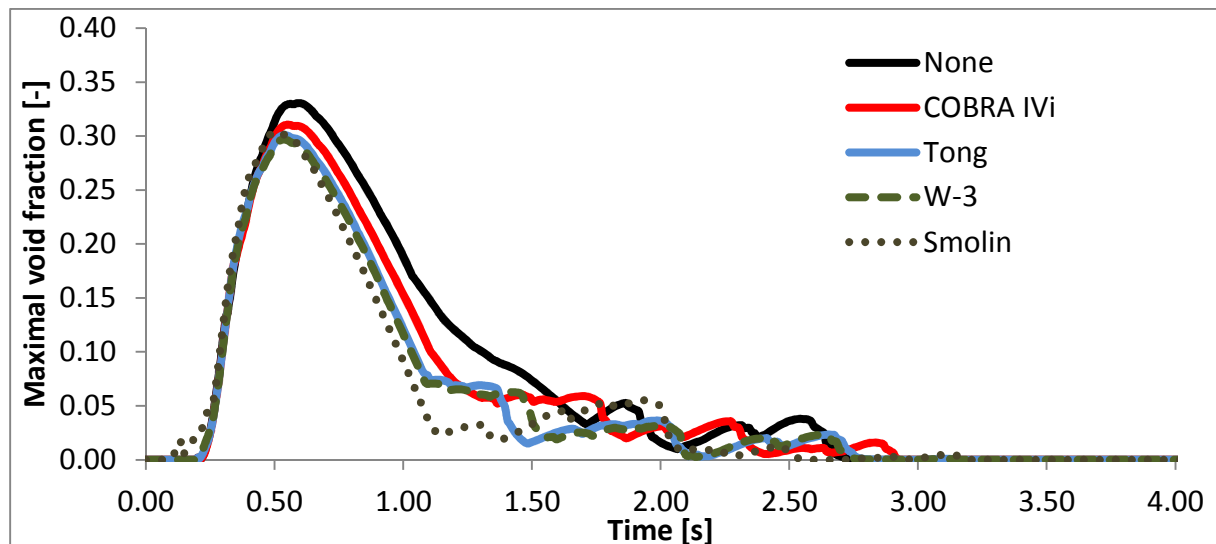


Fig. 10 Maximal values of void fraction for all shapes with W-3 CHF correlation

In addition, the evaluation of the maximal values, it is important to find the location of the maximum values. The location of the hottest pin (in terms of superheating) in the central assembly during the transient is given in Fig. 11 and in almost all cases and during almost the complete time of the transient the pin 163 is identified as the hottest one. The only exception is when the Smolin form factor is used and pin no. 127 is identified. Because of the symmetry of the FA pin 163 and 127 are identical. The differences arise only due to numerical issues in the last digits of the calculated values. In general, values four central pins are identical. The axial location of the maximum in the hottest pin (again, in terms of superheating) is given in Fig. 12. It indicates that during the first two seconds of the transient the maximum occurs below the mid-plane while during the last seconds of the transient the maximum moves to higher elevations due to the moving of the control rods. But it has also a wider spread, between 1.4 and 2.0 m. As soon as the superheating returns to zero the pin no. and the axial position drop to zero, too.

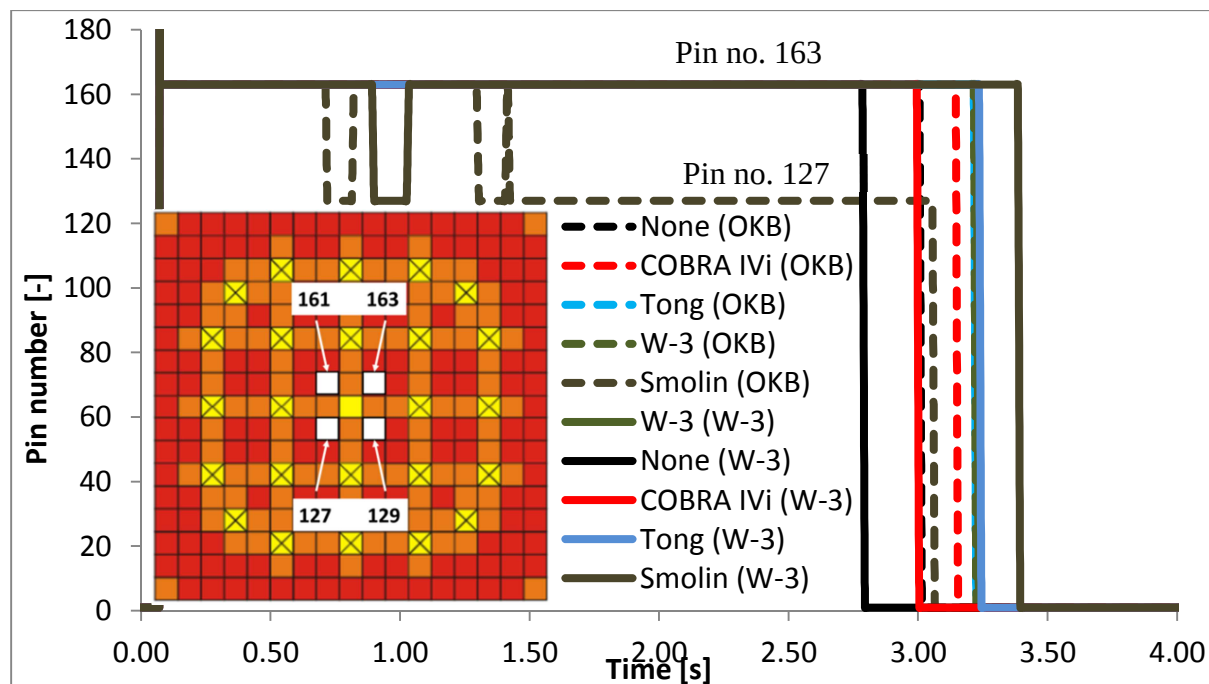


Fig. 11 Location of the hot pin during the transient

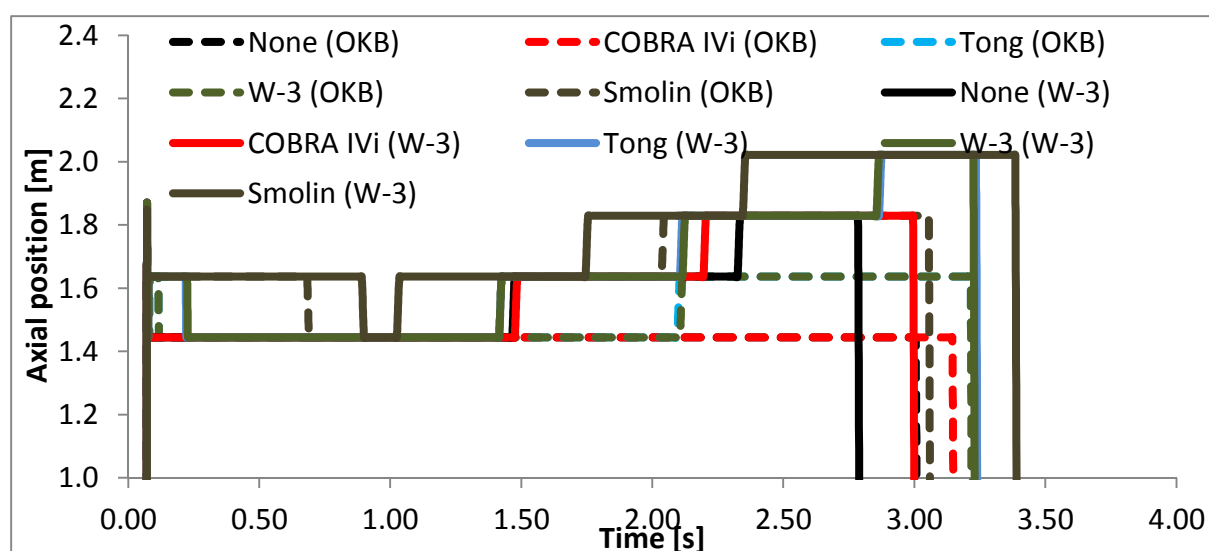


Fig. 12 Location of the hottest axial layer in the hot pin during the transient

Another finding of this investigation is that the position of the maximum void fraction does not correspond with the position of the maximal superheating/cladding temperature. Figure 13 shows on the left side the axial void fraction profile at 0.5 (top) and 1.0 (bottom) seconds, based on the W-3 CHF correlation with no form factor selected. It is clearly visible that the location of the maximum void fraction is at the corners of the assembly. The right side of Fig. 13 shows 2D plots of the void fraction at level 15 = 2.69 m (0.5 s) and level 20 = 3.66 m (1.0 s) and the corresponding mass flow rate distribution, which is lower at the corners than in the center of the FA. In addition, at the corners only fuel pins are present while in the center guide tubes or control rods are present (see Fig. 1), too. Due to the lower mass flow rate less energy is needed for the phase change. Besides the lower mass flow rate at the corners, the corner pins also experience superheating but not as pronounced as the center pins. These are the reason why the maximum void fraction occurs at the corners and not in the center.

In addition, the maximum of the void fraction is moving upwards with increasing transient time. The reason for that is that due to the control rod ejection the power density is moving, with delay, upwards. Thus, a maximum moves to higher elevations.

5. SUMMARY AND CONCLUSION

The aim of this paper is to demonstrate that the choice of CHF correlation and form factor has an impact on local safety parameters and should be considered prior to any similar analysis. In the present case the CHF models of OKB and W-3 are used together with a variety of form factors. The local thermal-hydraulic parameters vary but show a similar trend for all considered correlation and form factor options. With cladding temperatures of less than 800°C and maximum fuel centerline temperatures of 1500°C, acceptance criteria for accidents have not been exceeded and fuel integrity is maintained, meaning that no radioactivity is released as a consequence of the investigated transient. Even though a superheating of >400°C is calculated for the central pins the most void is produced at the corners of the FA.

It is shown that 4D power maps (time + space), generated with nodal neutronic codes and pin power reconstruction methods, is a mandatory prerequisite to identify phenomena like exceeding of critical heat flux, and boiling crisis at local scale.

The investigation demonstrated that the consideration of no correction measure for non-uniform heat fluxes yield similar results as it would be with form factors. This situation might change during other kinds of transient, especially once with a longer duration. The results with form factors show that none of these form factors is superior to the others. The one of Tong and the one of W-3 predict almost identical results.

Even though the considered transient (fresh fuel, ejection time = 0.1) is of unlikely and hypothetical nature and no experimental data are available to validate the simulation, the results are physically sound. The boiling crisis is of short nature and temperature limits are not exceeded. But a difference of 20°C in the superheating and cladding temperature between the two different CHF correlations indicates the need for further investigation. In the present transient the integral heat released in the system was too small to cause a total boiling or the higher cladding temperatures due to the short transient time. But during long lasting transient releasing huge amounts of energy the prediction using the one correlation could yield to a temperature limit violation while with the other one is still below the limit.

6. OUTLOOK

One task for future investigation is to address the influence of the axial nodalization. In the present study, 20 axial levels are modeled which will be increased to 40 or even 80 axial layers to meet the concerns related to REA transients.

Further improvements in the calculation can be obtained by means of high fidelity systems where the neutron transport equation is solved numerically at pin level and a two-way coupling approach is considered between DYN3D and SUBCHANFLOW [3]

Uncertainty and sensitivity studies will be performed in the future to support the decision making process concerning the selection of CHF correlations and form factors.

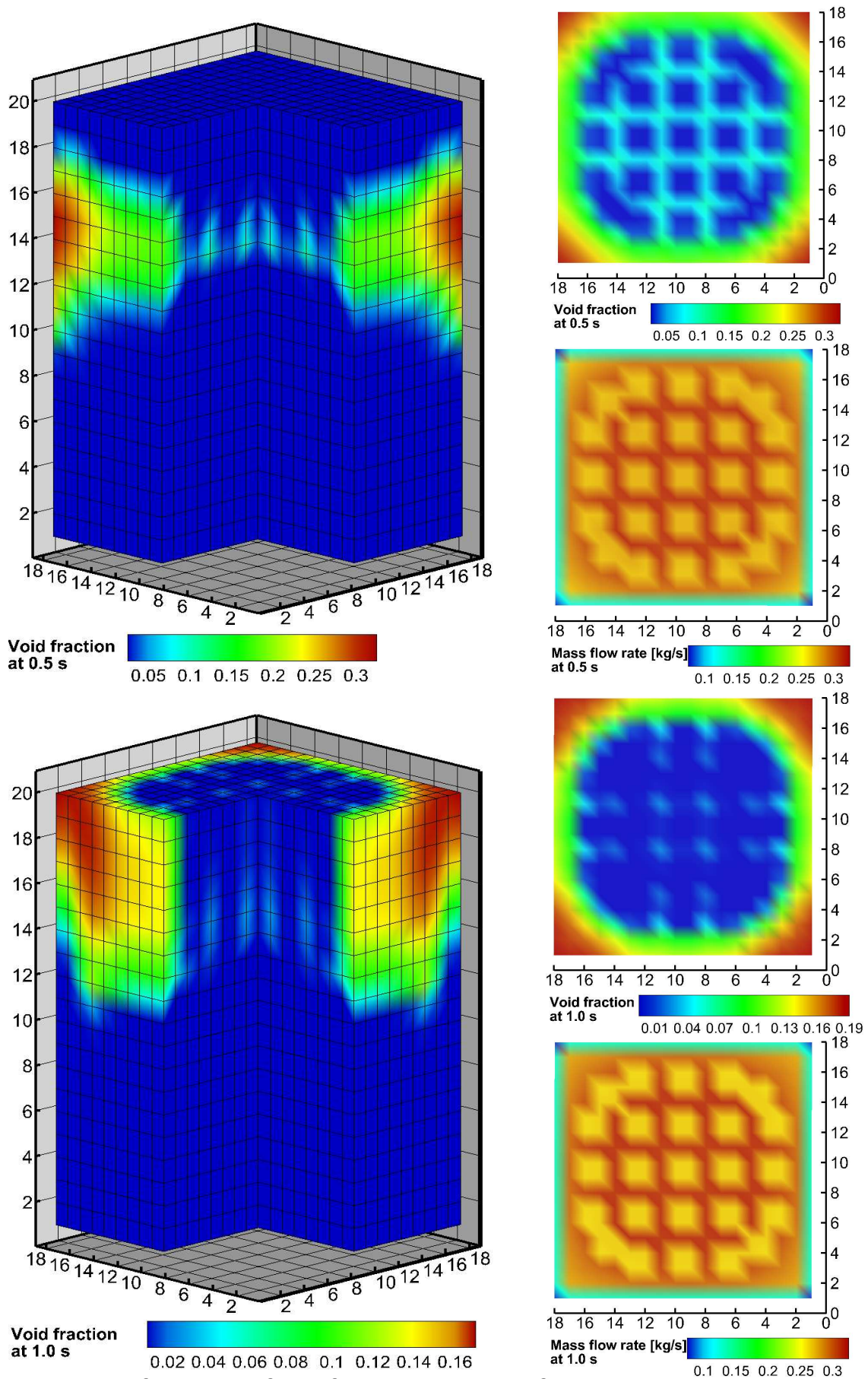


Fig. 13 3D void fraction profile (left side) and 2D void fraction and mass flow rate (right side) at 0.5 s and 2.69 m (top graphs) and 1.0 s and 3.66 m (bottom graphs)

NOMENCLATURE

Variables

α	Void fraction [-]
C	Form factor constant [-]
d	Hydraulic diameter [m]
F	Form factor for non-uniform heat flux profile [-]
G	Mass flux [kg/m ² s]
h	Enthalpy [kJ/kg]
l	Heated length [m]
MDNBR	Minimum Departure from Nucleate Boiling Ratio [-]
p	Pressure [Pa]
q''	Heat flux [W/m ²]
t	temperature [°C]
ΔT	Temperature difference [°C]
x	Steam quality [-]
Sub-script	
clad	Quantity of the cladding surface
cool	Quantity of the coolant bulk
crit	Quantity at critical point
DNB	Departure from Nucleate Boiling
fuel	Quantity of the fuel centerline
in	Inlet
local	Quantity at local point
sat	Saturation

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REFERENCES

- 1 C. Chauliac, "Overview on the Collaborative Project NURISP," *Presentation at the NURESIM General Seminar*, Madrid, Nov. 27-28, (2008).
- 2 U. Grundmann, S. Mittag, U. Rohde and S. Kliem, "DYN3D Version 3.2, Code for Calculation of Transients in Light Water Reactors (LWR), with Hexagonal or Quadratic Fuel Elements", Code Manual and Input Data Description for Release (2010).
- 3 A. Gomez, "Further developments of multiphysics and multiscale methodologies for coupled nuclear reactor simulations," PhD thesis at the Technische Universität München, (2011).
- 4 U. Imke, "SUBCHANFLOW: A new empirical knowledge based subchannel code," *Proceedings of Annual Meeting on Nuclear Technology*, Berlin, Germany (2010).
- 5 IAEA, "Accident Analysis for Nuclear Power Plants with Pressurized Water Reactors," Safety Report Series No. 30, IAEA (2003).
- 6 A. Gomez, G. Lenci, W. Jäger, V. Sánchez and K. Ivanov, "Light water reactor local safety parameter analysis using the reactor dynamics code DYN3D and

- SUBCHANFLOW,” *Proceedings 18th Pacific Basin Nuclear Conference (PBNC2012)*, Busan, Korea (2012).
- 7 S. Kliem, S. Mittag, A. Gommlich and P. Apanasevich, “*Definition of a boron dilution benchmark*,” Deliverable D3.1.2.2, NURISP Project, European Union (2011).
 - 8 T. Kozlowski, T. Downar, “*OECD/NEA and U.S. NRC PWR MOX/UO₂ core transient benchmark – Final specification*”, OECD Nuclear Energy Agency/Nuclear Science Committee (2003).
 - 9 V.P. Carey, “*Liquid-Vapor Phase-Change Phenomena*,” Hemisphere Publishing Corporation (1992).
 - 10 V.I. Astakhov, Y.A. Bezrukov, S.A. Logvinov and V.G. Brantov, “Analysis of the effect of the longitudinal heat release profile on burnout in rod bundles,” *Proceedings of Seminar on Thermophysical studies for ensuring reliability and safety of water-moderated water-cooled nuclear reactors*, Budapest, Hungary (1978).
 - 11 V.V. Bol’shakov, S.M. Bashkirtsev, L.L. Kobzar and A.G. Morozov, “Experimental study of burnout in channels with twisted fuel rods,” *Thermal Engineering*, **54** (5) pp. 386-389 (2007).
 - 12 L.S. Tong, “Prediction of departure from nucleate boiling for an axially non-uniform heat flux distribution,” *Journal of Nuclear Energy*, **21**, pp. 241-249 (1967).
 - 13 C.W. Stewart, C.L. Wheeler, R.J. Cena, C.A. McMonagle, J.M. Cuta and D.S. Trent, “*COBRA-IV: The model of the method*,” BNWL-2214, NRC-4 (1977).
 - 14 J.E. Kelly and M.S. Kazimi, “*Development and testing of the three dimensional, two-fluid code THERMIT for LWR core and subchannel applications*,” MIT Energy Laboratory Electric Utility Program, MIT-EL 79-046 (1979).
 - 15 V.N. Smolin and V.K. Polyakov, “Coolant boiling crisis in rod assemblies,” *Proceedings of 6th International Heat Transfer Conference*, Toronto, Canada (1978).